One constant per hydroxyl group: predicting freeze concentration and designing glycerol-sparing cryoprotectants

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Question. When cells freeze, ice is pure water, so salt concentrates in the liquid left behind. Ice–solution equilibrium fixes the water activity of that liquid at each temperature, $\ln(x_w\gamma_w) = -\frac{\Delta_{\mathrm{fus}}H}{R}\left(\frac{1}{T}-\frac{1}{T_m}\right) - \Delta C_p$ term, so the only unknown is γ_w. How little glycerol keeps the salt below the damage level, and how much of it can sorbitol replace?

Hypothesis (tested, not assumed). One water–OH interaction constant, $\ln\gamma_w = (1-x_w)\sum_i \frac{n_{\mathrm{OH},i}\,a_{\mathrm{OH}}}{RT}x_i$, fitted on glycerol only, predicts other polyols and polyol + salt mixtures without refitting.

How. Fit $a_{\mathrm{OH}} = -2415\,\mathrm{J\,mol^{-1}}$ on 19 CRC glycerol points (RMSE 0.017 K) → predictions committed to git → compared with independent literature data and five benchmark models → recipe. Everything rebuilds from one command (`run_all.py`).

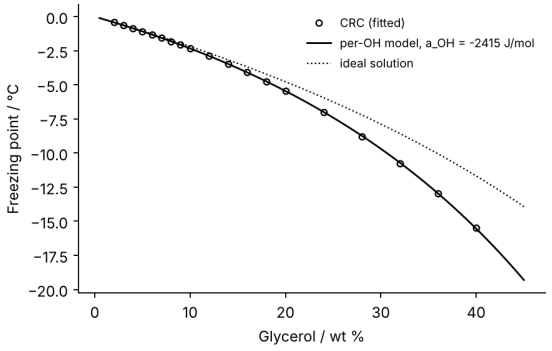


Fig. 1. The only fit: glycerol, one constant.

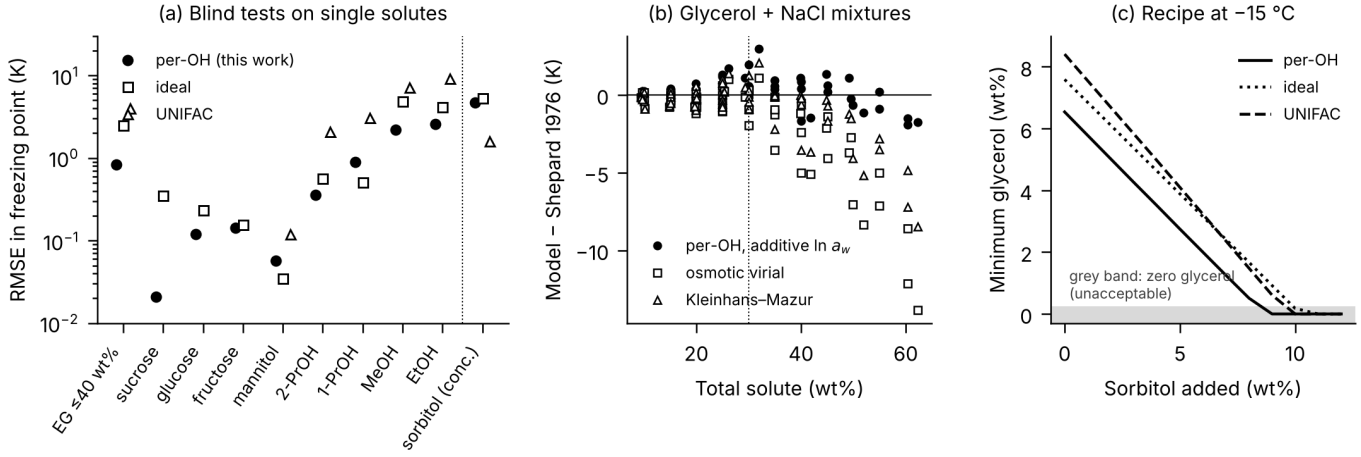


Fig. 2. (a) Blind single-solute tests (log scale). (b) Glycerol + NaCl liquidus (Shepard 1976): our additive rule vs the two cryobiology models. (c) Glycerol needed vs sorbitol added, three activity models, −15 °C, NaCl ≤ 4.41 wt%.

Results so far

- **Transfers:** ethylene glycol 0.83 K (ideal 2.49, UNIFAC 3.99); sucrose 0.02 K (ideal 0.35).
- **Mixtures:** glycerol + NaCl above 30 wt% solute 1.23 K vs osmotic virial 5.75 and Kleinhans–Mazur 3.39. Below 30 wt%: 0.69 K, a narrow miss of our pre-set 0.5 K target.
- **Fails:** sorbitol (concentrated, and at 25 °C) and monohydric alcohols. A sucrose two-temperature test shows the sorbitol failure is the molecule, not temperature.
- **UNIFAC** predicts the wrong sign of non-ideality for water–glycerol.
- **Design:** 1 g sorbitol saves 0.68–0.91 g glycerol across 3 models × 3 temperatures × 3 salt limits.
- **Effort note:** Shepard (1976) gives no tables; its Fig. 3 was digitised by hand (76 points, checked against the NaCl–water eutectic; figure accuracy measured at 0.34 K against CRC).

Next (before 6 Nov)

- Lab: 5 committed samples in duplicate; expected $U \approx 0.25$ K vs model spreads 0.7–6.6 K.
- UNIFAC cross-check in DWSIM; post-hoc per-molecule fit, labelled as such.

Feedback I would like

1. Is reporting the failures (sorbitol, alcohols, the 0.5 K miss) as results the right framing for this proposal?
2. Lab: dates; is a conductivity meter or refractometer available to measure salt in the unfrozen liquid?
3. Is a post-hoc section fine in the 5 pages, or should it be an appendix?
4. May the 5 pages use a journal-paper layout, keeping your section order?
5. Aspen Plus access for the UNIFAC check?